SG No. 53 D_{2h}^{7} *Pmna* [orthorhombic]

* plus set: $+[0, 0, 0]$

* Wyckoff site: `2a`, site symmetry: `2/m..`

Table 1: Wyckoff bond: `2a@2a`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$[0, 0, 0]$	`[1,-4,-5,8]`
2	$[0, -Y, Z]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[2,-3,-6,7]`

Table 2: Wyckoff bond: `2b@2a`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[0, 0, 0]$	`[1,4,-5,-8]`
2	$[-X, 0, 0]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[2,3,-6,-7]`

Table 3: Wyckoff bond: `4c@2a`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, 0]$	`[1,-5]`
2	$[-X, -Y, Z]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[2,-6]`
3	$[-X, Y, -Z]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[3,-7]`
4	$[X, -Y, -Z]$	$[0, 0, 0]$	`[4,-8]`

* Wyckoff site: `2b`, site symmetry: `2/m..`

Table 4: Wyckoff bond: `2a@2b`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$[\frac{1}{2}, 0, 0]$	`[1,-4,-5,8]`
2	$[0, -Y, Z]$	$[0, 0, \frac{1}{2}]$	`[2,-3,-6,7]`

Table 5: Wyckoff bond: `2b@2b`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[\frac{1}{2}, 0, 0]$	`[1,4,-5,-8]`
2	$[-X, 0, 0]$	$[0, 0, \frac{1}{2}]$	`[2,3,-6,-7]`

Table 6: Wyckoff bond: `4c@2b`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{1}{2}, 0, 0]$	`[1,-5]`
2	$[-X, -Y, Z]$	$[0, 0, \frac{1}{2}]$	`[2,-6]`
3	$[-X, Y, -Z]$	$[0, 0, \frac{1}{2}]$	`[3,-7]`

4	$[X, -Y, -Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[4,-8]`

* Wyckoff site: `2c`, site symmetry: `2/m..`

Table 7: Wyckoff bond: `2a@2c`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,-4,-5,8]`
2	$[0, -Y, Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-3,-6,7]`

Table 8: Wyckoff bond: `2b@2c`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,4,-5,-8]`
2	$[-X, 0, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2,3,-6,-7]`

Table 9: Wyckoff bond: `4c@2c`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,-5]`
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-6]`
3	$[-X, Y, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,-7]`
4	$[X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[4,-8]`

* Wyckoff site: `2d`, site symmetry: `2/m..`

Table 10: Wyckoff bond: `2a@2d`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[1,-4,-5,8]`
2	$[0, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-3,-6,7]`

Table 11: Wyckoff bond: `2b@2d`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[1,4,-5,-8]`
2	$[-X, 0, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,3,-6,-7]`

Table 12: Wyckoff bond: `4c@2d`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[1,-5]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-6]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[3,-7]`

continued ...

Table 12

No.	vector	center	mapping
4	$[X, -Y, -Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[4,-8]`

* Wyckoff site: `4e`, site symmetry: `2..`

Table 13: Wyckoff bond: `4a@4e`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$[x, 0, 0]$	`[1,-4]`
2	$[0, -Y, Z]$	$\left[\frac{1}{2} - x, 0, \frac{1}{2}\right]$	`[2,-3]`
3	$[0, -Y, -Z]$	$[-x, 0, 0]$	`[5,-8]`
4	$[0, Y, -Z]$	$\left[x + \frac{1}{2}, 0, \frac{1}{2}\right]$	`[6,-7]`

Table 14: Wyckoff bond: `4b@4e`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[x, 0, 0]$	`[1,4]`
2	$[-X, 0, 0]$	$\left[\frac{1}{2} - x, 0, \frac{1}{2}\right]$	`[2,3]`
3	$[-X, 0, 0]$	$[-x, 0, 0]$	`[5,8]`
4	$[X, 0, 0]$	$\left[x + \frac{1}{2}, 0, \frac{1}{2}\right]$	`[6,7]`

Table 15: Wyckoff bond: `8c@4e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, 0, 0]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2} - x, 0, \frac{1}{2}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{2} - x, 0, \frac{1}{2}\right]$	`[3]`
4	$[X, -Y, -Z]$	$[x, 0, 0]$	`[4]`
5	$[-X, -Y, -Z]$	$[-x, 0, 0]$	`[5]`
6	$[X, Y, -Z]$	$\left[x + \frac{1}{2}, 0, \frac{1}{2}\right]$	`[6]`
7	$[X, -Y, Z]$	$\left[x + \frac{1}{2}, 0, \frac{1}{2}\right]$	`[7]`
8	$[-X, Y, Z]$	$[-x, 0, 0]$	`[8]`

* Wyckoff site: `4f`, site symmetry: `2..`

Table 16: Wyckoff bond: `4a@4f`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$\left[x, \frac{1}{2}, 0\right]$	`[1,-4]`
2	$[0, -Y, Z]$	$\left[\frac{1}{2} - x, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-3]`
3	$[0, -Y, -Z]$	$\left[-x, \frac{1}{2}, 0\right]$	`[5,-8]`
4	$[0, Y, -Z]$	$\left[x + \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[6,-7]`

Table 17: Wyckoff bond: `4b@4f`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$\left[x,\ \frac{1}{2},\ 0\right]$	`[1,4]`
2	$[-X,\ 0,\ 0]$	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,3]`
3	$[-X,\ 0,\ 0]$	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[5,8]`
4	$[X,\ 0,\ 0]$	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[6,7]`

Table 18: Wyckoff bond: `8c@4f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[x,\ \frac{1}{2},\ 0\right]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2]`
3	$[-X,\ Y,\ -Z]$	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3]`
4	$[X,\ -Y,\ -Z]$	$\left[x,\ \frac{1}{2},\ 0\right]$	`[4]`
5	$[-X,\ -Y,\ -Z]$	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[5]`
6	$[X,\ Y,\ -Z]$	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[6]`
7	$[X,\ -Y,\ Z]$	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[7]`
8	$[-X,\ Y,\ Z]$	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[8]`

* Wyckoff site: `4g`, site symmetry: `.2.`

Table 19: Wyckoff bond: `4a@4g`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X,\ 0,\ Z]$	$\left[\frac{1}{4},\ y,\ \frac{1}{4}\right]$	`[1,-3]`
2	$[-X,\ 0,\ Z]$	$\left[\frac{1}{4},\ -y,\ \frac{3}{4}\right]$	`[2,-4]`
3	$[-X,\ 0,\ -Z]$	$\left[\frac{3}{4},\ -y,\ \frac{3}{4}\right]$	`[5,-7]`
4	$[X,\ 0,\ -Z]$	$\left[\frac{3}{4},\ y,\ \frac{1}{4}\right]$	`[6,-8]`

Table 20: Wyckoff bond: `4b@4g`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[0,\ Y,\ 0]$	$\left[\frac{1}{4},\ y,\ \frac{1}{4}\right]$	`[1,3]`
2	$[0,\ -Y,\ 0]$	$\left[\frac{1}{4},\ -y,\ \frac{3}{4}\right]$	`[2,4]`
3	$[0,\ -Y,\ 0]$	$\left[\frac{3}{4},\ -y,\ \frac{3}{4}\right]$	`[5,7]`
4	$[0,\ Y,\ 0]$	$\left[\frac{3}{4},\ y,\ \frac{1}{4}\right]$	`[6,8]`

Table 21: Wyckoff bond: `8c@4g`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{4},\ y,\ \frac{1}{4}\right]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$\left[\frac{1}{4},\ -y,\ \frac{3}{4}\right]$	`[2]`
3	$[-X,\ Y,\ -Z]$	$\left[\frac{1}{4},\ y,\ \frac{1}{4}\right]$	`[3]`
4	$[X,\ -Y,\ -Z]$	$\left[\frac{1}{4},\ -y,\ \frac{3}{4}\right]$	`[4]`
5	$[-X,\ -Y,\ -Z]$	$\left[\frac{3}{4},\ -y,\ \frac{3}{4}\right]$	`[5]`

continued ...

Table 21

No.	vector	center	mapping
6	$[X, Y, -Z]$	$\left[\frac{3}{4}, y, \frac{1}{4}\right]$	`[6]`
7	$[X, -Y, Z]$	$\left[\frac{3}{4}, -y, \frac{3}{4}\right]$	`[7]`
8	$[-X, Y, Z]$	$\left[\frac{3}{4}, y, \frac{1}{4}\right]$	`[8]`

* Wyckoff site: `4h`, site symmetry: `m..`

Table 22: Wyckoff bond: `4a@4h`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$[0, y, z]$	`[1,8]`
2	$[0, -Y, Z]$	$\left[\frac{1}{2}, -y, z+\frac{1}{2}\right]$	`[2,7]`
3	$[0, Y, -Z]$	$\left[\frac{1}{2}, y, \frac{1}{2}-z\right]$	`[3,6]`
4	$[0, -Y, -Z]$	$[0, -y, -z]$	`[4,5]`

Table 23: Wyckoff bond: `4b@4h`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[0, y, z]$	`[1,-8]`
2	$[-X, 0, 0]$	$\left[\frac{1}{2}, -y, z+\frac{1}{2}\right]$	`[2,-7]`
3	$[-X, 0, 0]$	$\left[\frac{1}{2}, y, \frac{1}{2}-z\right]$	`[3,-6]`
4	$[X, 0, 0]$	$[0, -y, -z]$	`[4,-5]`

Table 24: Wyckoff bond: `8c@4h`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, y, z]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, -y, z+\frac{1}{2}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}, y, \frac{1}{2}-z\right]$	`[3]`
4	$[X, -Y, -Z]$	$[0, -y, -z]$	`[4]`
5	$[-X, -Y, -Z]$	$[0, -y, -z]$	`[5]`
6	$[X, Y, -Z]$	$\left[\frac{1}{2}, y, \frac{1}{2}-z\right]$	`[6]`
7	$[X, -Y, Z]$	$\left[\frac{1}{2}, -y, z+\frac{1}{2}\right]$	`[7]`
8	$[-X, Y, Z]$	$[0, y, z]$	`[8]`

* Wyckoff site: `8i`, site symmetry: `1`

Table 25: Wyckoff bond: `8a@8i`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}-x, -y, z+\frac{1}{2}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}-x, y, \frac{1}{2}-z\right]$	`[3]`
4	$[X, -Y, -Z]$	$[x, -y, -z]$	`[4]`
5	$[-X, -Y, -Z]$	$[-x, -y, -z]$	`[5]`
6	$[X, Y, -Z]$	$\left[x+\frac{1}{2}, y, \frac{1}{2}-z\right]$	`[6]`

continued ...

Table 25

No.	vector	center	mapping
7	$[X, -Y, Z]$	$\left[x+\frac{1}{2}, -y, z+\frac{1}{2}\right]$	[7]
8	$[-X, Y, Z]$	$[-x, y, z]$	[8]